LESSON PLAN · PYTHON (GR 6–8) INTERMEDIATE: PYTHON GAME CODERS · LEVEL 2

Lesson 2: Turtle Graphics

Module 1 | 60 minutes | turtle, loops, functions | Grades 7–11 · ages 12–16

Objective: Draw shapes and patterns with the turtle module, using loops and functions with parameters, and build a Diwali rangoli generator.

Success criteria

- I can move and turn the turtle to draw lines.
- I can work out the turn angle for any shape (360 ÷ sides).
- I can write a function with parameters that draws a shape.
- I can repeat and rotate a shape to make a pattern.

Key vocabulary

- **turtle**: A drawing robot in Python that moves and draws lines.
- **forward(n)**: Moves the turtle n steps, drawing a line if the pen is down.
- **right(a) / left(a)**: Turns the turtle by a degrees.
- **angle**: How far to turn, measured in degrees; a full turn is 360.
- **penup / pendown**: Lift or lower the pen so moving draws or does not draw.
- **polygon**: A closed shape with straight sides, like a triangle or hexagon.

You will need

- Projector with Kitabash (lesson video, Python Lab)
- Devices: 1 per learner, with a keyboard
- Printed worksheet (optional)
- Project files: turtle_shapes.py

Timed lesson plan (60 min)

Time	Stage	What happens
5 min	**Warm-up game: Human Turtle**	Play Human Turtle (unplugged): learners steer a classmate (or the teacher's cursor) into a square and a triangle with forward/turn commands.
5 min	**Hook and video**	Show the hook. Poll: "What shape will 4 forwards and 4 right turns make?" Play the video with captions; pause at "One function, every shape".
10 min	**Live demo and build-along**	Run the live demo (including the wrong 60 degree triangle), then everyone types the shapes build-along. Thumbs check: "Do you see three shapes?"

8 min	**Challenge and angle game**	Any-shape function challenge, the predict-the-drawing question and the angle match. Quick competition: first to post the turn angle for a 10-sided shape (36) wins.
15 min	**Rangoli project in breakout pairs**	In breakout pairs, learners plan their rangoli on paper (shape, number of petals, colours) and code it. Partners suggest one change to try.
6 min	**Debug race**	Race to fix “the square that ran away”. The winner explains why indentation matters.
7 min	**Rangoli gallery show-and-tell**	Learners share screenshots in a class gallery; three volunteers explain their angle and colours. Vote for the most colourful with emoji reactions.
4 min	**Wrap-up and exit ticket**	Thumbs check on each success criterion. Exit ticket: the octagon angle.

Unplugged warm-up

Human Turtle (5 min game): one learner is the turtle and stands up; classmates give only “forward 3 steps” and “turn right 90” commands to make them walk a square, then a triangle. Online: the teacher moves a cursor on a whiteboard as learners type commands in the chat. Who can make a triangle in the fewest commands?

Teaching the idea

The turtle module lets Python draw: forward draws a line, right and left turn by an angle in degrees. A closed shape always turns a total of 360 degrees, so each corner of an n-sided shape turns 360 ÷ n. A loop repeats forward-and-turn; wrapping it in a function with parameters (sides, size) lets one line draw any shape. Patterns like rangoli come from repeating a shape and rotating a little each time.

Analogy: The turtle is like a remote-control car with a pen taped underneath: it only knows “drive forward” and “turn”, but clever sequences of those two commands can draw anything.

Live demo: 1) Type import turtle and t = turtle.Turtle(), then t.forward(100); run and point to the canvas. 2) Add t.right(90) and t.forward(100); ask the class which way it will go before running. 3) Type the square with a for loop. 4) Ask: “What angle for a triangle?” Try 60 first (wrong!) and run to show the open shape, then 120. 5) Wrap it into def draw_polygon(sides, size): and call it for 3, 5 and 8 sides. 6) Show the rangoli loop: draw_polygon(6, 60) and t.right(30) inside for i in range(12).

Questions to ask

Ask	Listen for
What does t.forward(100) do?	Moves the turtle 100 steps and draws a line.
What angle do you turn for a pentagon?	72 degrees, because 360 ÷ 5 = 72.

Why put the drawing loop in a function?	So one line can draw any shape with different sides and sizes.
How do you move without drawing?	t.penup(), move, then t.pendown().
How does a rangoli pattern repeat?	Draw a shape, turn a little, and repeat until you have turned 360 degrees.

Differentiation

Support
Give a turn-angle table (3 → 120, 4 → 90, 5 → 72, 6 → 60, 8 → 45). Let learners start from the build-along file and only change the numbers and colours.

Stretch
Make the size grow inside the loop (size = size + 5) to draw a spiral, or draw a star with 5 sides and a 144 degree turn.

Common misconceptions

Misconception	How to fix it
A triangle turns 60 degrees	The turtle turns the outside angle: 120 degrees for a triangle.
right(90) moves the turtle	Turning happens on the spot. Only forward and backward move it.
Code after the loop repeats too	Only the indented lines repeat.
The function draws as soon as you define it	Nothing is drawn until you call draw_polygon(...).

Assessment

Use the Python Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
What angle should the turtle turn at each corner of an octagon, and why?

Homework
Make a rangoli card for someone at home: use their favourite colours and add your own extra layer. Ask them to choose the number of petals.

Answer key — Kitabash activities

1. Python Lab: any-shape function (perform · python lab)

Goals checked: Draws a closed hexagon · Calls draw_polygon · Uses a function, a loop and turtle

One solution:

```
import turtle
t = turtle.Turtle()

def draw_polygon(sides, size):
    for i in range(sides):
        t.forward(size)
        t.right(360 / sides)

draw_polygon(6, 80)
```

2. Predict the drawing: three turns (perform · code output)

3. Match each shape to the turn at every corner (perform · matching)

Triangle → 120 degrees · Square → 90 degrees · Pentagon → 72 degrees · Hexagon → 60 degrees · Octagon → 45 degrees

4. Project: Diwali rangoli generator (perform · python lab)

Goals checked: Draws a big pattern · Repeats the shape function · Uses functions, loops and turtle

Example program:

```
# Diwali rangoli generator
# Made by: Meera
import turtle
t = turtle.Turtle()
t.speed(0)
turtle.bgcolor("black")

colours = ["orange", "magenta", "yellow", "cyan"]

def draw_polygon(sides, size):
    for i in range(sides):
        t.forward(size)
        t.right(360 / sides)

def petal_ring(count, size):
    for i in range(count):
        t.color(colours[i % len(colours)])
        draw_polygon(6, size)
        t.right(360 / count)

def diya_dots(count, radius):
    for i in range(count):
        t.penup()
        t.forward(radius)
        t.dot(10, "gold")
        t.backward(radius)
        t.pendown()
        t.right(360 / count)

petal_ring(12, 60)
petal_ring(8, 30)
diya_dots(12, 130)
t.hideturtle()
```

5. Debug it: the square that ran away (probe · python lab)

Goals checked: Draws 4 sides · The square is closed

One solution:

```
import turtle
t = turtle.Turtle()
t.color("orange")
for i in range(4):
    t.forward(100)
    t.right(90)
```

Answer key — worksheet

1. **Key words:** turtle = A drawing robot in Python that moves and draws lines. · forward(n) = Moves the turtle n steps, drawing a line if the pen is down. · right(a) / left(a) = Turns the turtle by a degrees. · angle = How far to turn, measured in degrees; a full turn is 360.
2. **Plan your program:** Goals: Draws a closed hexagon; Calls draw_polygon; Uses a function, a loop and turtle. One solution: import turtle t = turtle.Turtle() def draw_polygon(sides, size): for i in range(sides): t.forward(size) t.right(360 / sides) draw_polygon(6, 80)
3. **Predict the drawing: three turns:** A. A triangle
4. **Match each shape to the turn at every corner:** Triangle → 120 degrees · Square → 90 degrees · Pentagon → 72 degrees · Hexagon → 60 degrees · Octagon → 45 degrees
5. **Plan your program:** Goals: Draws 4 sides; The square is closed. One solution: import turtle t = turtle.Turtle() t.color("orange") for i in range(4): t.forward(100) t.right(90)

Kitabash original curriculum.